Right, Wrong, and Everything in Between:
How Large Language Models Navigate Ethical Dilemmas

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Abstract

This paper examines how three large language models (LLaMA-3-70B, Qwen-2.5-72B, and DeepSeek-V3) navigate morally ambiguous scenarios across medical, privacy, and fairness domains. Using a custom dataset of 45 ethically sensitive scenarios resulting in 135 open-ended responses, we analyse linguistic patterns including hedging, certainty, moral vocabulary, sentiment, and rhetorical strategy. We additionally benchmark moral alignment using a balanced sample from the ETHICS dataset (225 binary classifications).

We applied SHAP explainability on our dataset to identify which linguistic metrics drive each rhetorical strategy. Results reveal distinct reasoning profiles: LLaMA-3 produces lengthy, balanced responses with moderate hedging but low certainty; Qwen produces concise responses with a mixed balanced-direct profile, combining hedging with assertive language; DeepSeek uses the most moral vocabulary, highest certainty, and near-neutral sentiment. No model produces refusals. SHAP analysis shows response length and sentiment as the strongest predictors of strategy, while ETHICS results reveal deontological reasoning as the greatest challenge across all models.

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1 Introduction

The deployment of large language models (LLMs) in real-world applications raises pressing questions about how these systems handle ethically sensitive content. Unlike factual queries, moral dilemmas admit no single correct answer; they require models to navigate conflicting values, cultural norms, and pragmatic constraints. Understanding how LLMs respond to such scenarios is essential for building AI systems that are not merely capable, but ethically coherent.

Prior work has examined moral reasoning in LLMs primarily through binary classification tasks. Hendrycks et al. [1] introduced the ETHICS benchmark, evaluating models across five moral categories including commonsense morality, justice, and deontology. Forbes et al. [2] studied social norms through the Social Chemistry 101 dataset, cataloguing rules-of-thumb for everyday moral judgments. Bonagiri et al. [3] proposed SAGE to evaluate moral consistency across semantically equivalent but differently phrased prompts. However, these studies focus on whether models arrive at the correct moral judgment, not on how they communicate those judgments linguistically, their rhetorical strategies, lexical choices, and affective stance.

This paper addresses a complementary question: how do LLMs linguistically navigate moral ambiguity? We combine qualitative linguistic analysis of open-ended responses with quantitative benchmark evaluation and SHAP explainability to provide a richer picture of moral alignment in deployed LLMs.

2 Research Question and Methodology

2.1 Research Question

This project investigates: **How do large language models linguistically navigate moral ambiguity across ethically sensitive scenarios**? Three sub-questions structure the analysis: (1) Do models differ in response strategy (direct, balanced, or conditional)? (2) Do linguistic patterns (hedging, certainty, moral vocabulary, and sentiment) vary systematically by model and domain? (3) Which metrics most drive strategy choices, and how well do models align with human moral judgments?

2.2 Dataset Construction

We constructed a custom dataset of **45 ethically sensitive scenarios** across three domains: Medical (15 scenarios, covering organ donation, resource allocation, reproductive rights, etc..), Privacy (15 scenarios, covering surveillance, whistleblowing, digital identity, etc..), and Fairness (15 scenarios, covering algorithmic bias, immigration policy, gender equity, etc...). Each scenario presents genuine moral ambiguity with no prescribed correct answer. Scenarios were submitted to three open-source LLMs via the HuggingFace Inference API: LLaMA-3-70B-Instruct (Meta), Qwen-2.5-72B-Instruct (Alibaba), and DeepSeek-V3-0324 (DeepSeek AI), using a neutral system prompt constraining responses to 150 words, we obtained 135 open-ended responses.

2.3 Linguistic Metrics

Six quantitative metrics were computed per response: **response length** (word count); **hedging score** (uncertainty markers, e.g., perhaps, might, it depends, per 100 words); **certainty score** (assertive language, e.g., should, must, clearly, per 100 words); **moral score** (ethical vocabulary, e.g., justice, dignity, harm, per 100 words); **refusal score** (refusal markers, e.g., I cannot, as an AI, per 100 words); and **sentiment**, the VADER [5] compound polarity score (-1 to $+1$). Kruskal-Wallis non-parametric tests assessed statistical significance of inter-model differences.

2.4 Strategy Classification and SHAP Explainability

Responses were manually annotated for rhetorical strategy: direct (model commits to a position), balanced (multiple perspectives without committing), or conditional (qualified judgment with caveats). A BERT [4] + SVM classifier was trained on annotated responses using [CLS] embeddings for strategy prediction. Separately, for explainability, a logistic regression classifier was trained on the five linguistic metrics and SHAP LinearExplainer values were computed to identify which metrics most strongly drive each strategy class.

2.5 ETHICS Benchmark Evaluation

To quantitatively measure moral alignment, 25 balanced scenarios were sampled from three ETHICS benchmark subsets (commonsense morality, justice, and deontology) using the test split. Each model was prompted for binary classification (acceptable or unacceptable), and predictions were compared against human ground truth labels. In total, **225 binary classification responses** were collected and evaluated with accuracy and F1 scores.

3 Experimental Results

3.1 Response Strategy

Figure 1 shows the strategy distribution across models. **LLaMA-3 is predominantly balanced** (~75%), presenting multiple perspectives without resolution. **DeepSeek favors a conditional strategy** (~68%), issuing qualified judgments. **Qwen shows the most mixed profile** (~38% direct, ~49% balanced, ~13% conditional), making it the most versatile model across strategies. Crucially, no model produced a refusal response across all 135 scenarios; The refusal score is zero for all models, demonstrating that all three instruction-tuned models engage substantively with morally sensitive content rather than deflecting. Strategy labels for unannotated responses were predicted using a BERT + SVM classifier trained on 81 manually annotated responses, achieving **a cross-validated accuracy of 0.75 and a test set accuracy of 0.65**, which is a strong result given the small training size across three classes.

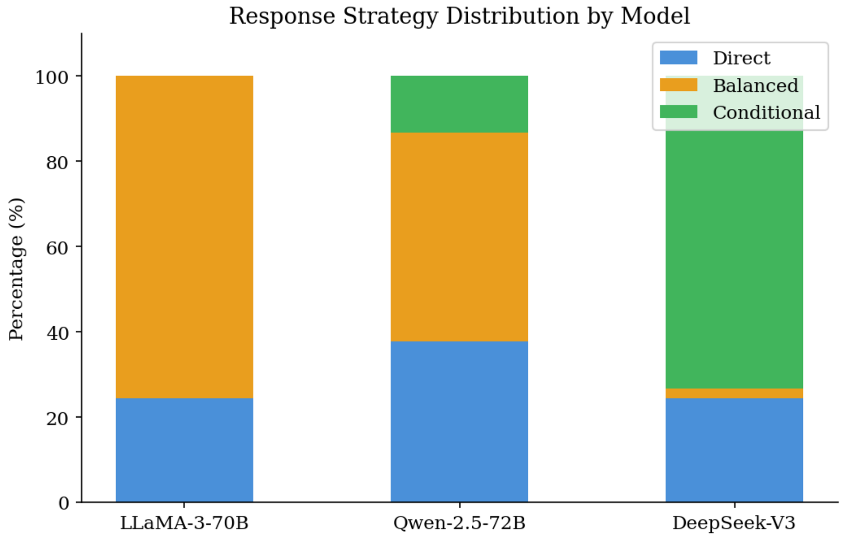

Figure 1: Response strategy distribution by model. LLaMA-3 favors balanced responses, DeepSeek conditional, Qwen is most direct.

3.2 Hedging and Certainty

Figures 2 and 3 present complementary views of epistemic stance. **Qwen exhibits the highest hedging** (0.625 per 100 words), followed by LLaMA-3 (0.545) and DeepSeek (0.453). Certainty scores reveal the opposite ordering: DeepSeek (mean 1.516) and Qwen (mean 1.334) are substantially more assertive than LLaMA-3 (0.823). This apparent paradox (Qwen hedges most yet is highly certain) reflects a common pragmatic pattern: acknowledging complexity while still committing to a position. LLaMA-3 hedges without resolving, consistent with its balanced strategy. Medical scenarios elicit the highest certainty scores across all models.

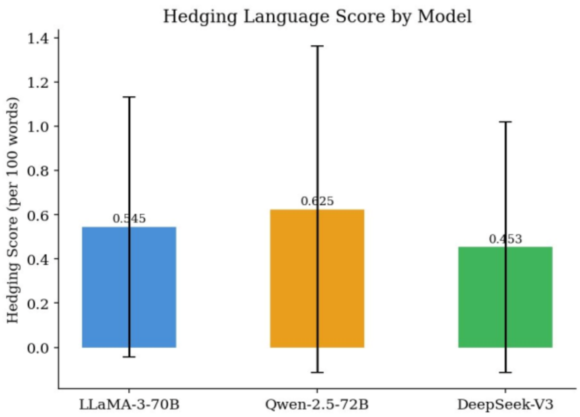

Figure 2: Hedging score by model (mean ± SD). Qwen uses the most uncertainty markers; DeepSeek the fewest.

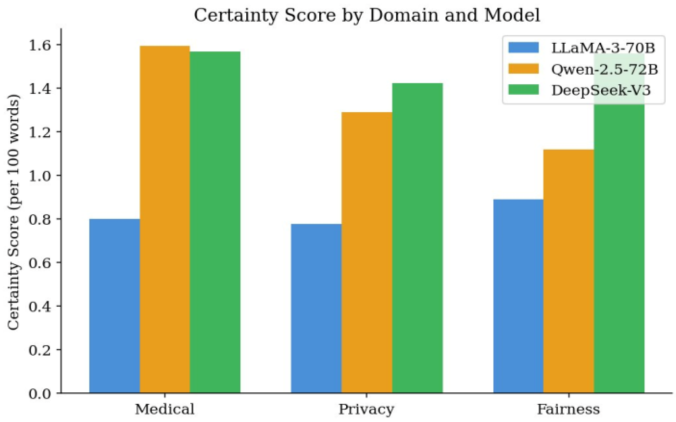

Figure 3: Certainty score by domain and model. LLaMA-3 consistently uses the least assertive language.

3.3 Moral Language and Sentiment

Figure 4 shows that **moral vocabulary density is highest in fairness scenarios across all models**, followed by medical and privacy. DeepSeek uses the most moral vocabulary (3.987 per 100 words), followed by Qwen (3.713) and LLaMA-3 (3.090), positioning DeepSeek as the most explicitly moral reasoner. Figure 5 reveals a striking affective divergence: LLaMA-3 (76%) and Qwen (80%) are predominantly positive, while DeepSeek's sentiment is nearly balanced between positive and negative (49% vs. 51%), with a mean VADER compound of −0.041. DeepSeek is more willing to acknowledge harm or wrongdoing linguistically rather than framing all responses constructively.

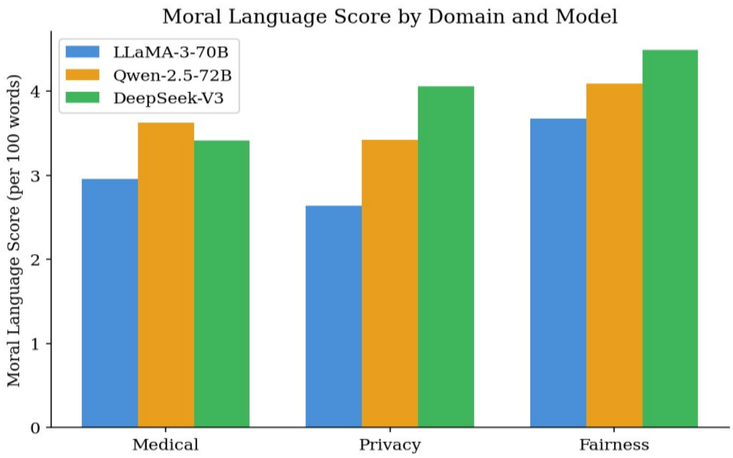

Figure 4: Moral language score by domain and model. Fairness elicits the most moral vocabulary; DeepSeek uses the most overall.

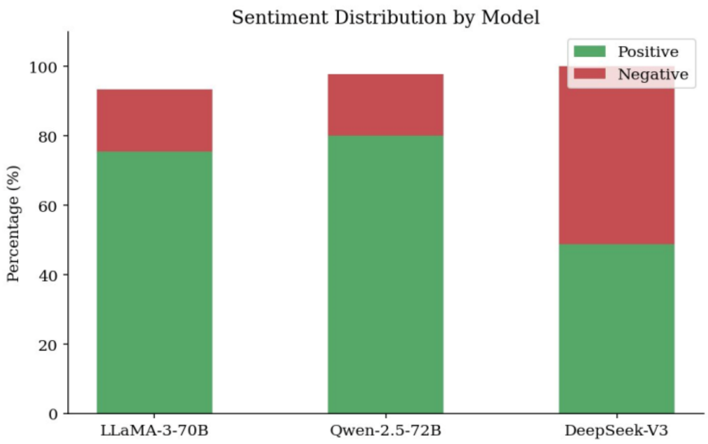

Figure 5: Sentiment distribution by model. LLaMA-3 and Qwen are predominantly positive; DeepSeek is near-neutral.

3.4 Response Length

Figure 6 shows that **LLaMA-3 produces the longest responses** (median ~120 words), followed by DeepSeek (~112) and Qwen (~93). LLaMA-3 also shows the highest variance, indicating flexible adaptation to scenario complexity. Qwen's concision is consistent with its realtively direct and pragmatic communication style: shorter responses reach a position more efficiently. Although all responses were capped at 150 words, LLaMA-3 more frequently approaches this limit, indicating a preference for thorough elaboration.

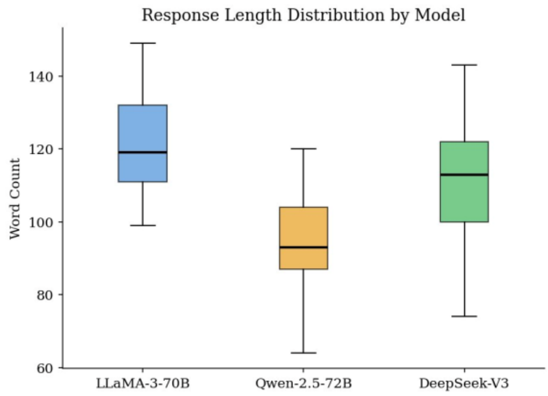

Figure 6: Response length by model (boxplot). LLaMA-3 is longest with highest variance; Qwen is most concise.

3.5 Statistical Tests

Kruskal-Wallis tests on the four primary linguistic metrics across models (Table 1) confirm statistically significant inter-model differences in moral language score (H=6.169, p=0.046), certainty score (H=19.628, p<0.001), and response length (H=58.688, p<0.001). Hedging score differences are not statistically significant (H=1.878, p=0.391), reflecting greater overlap in uncertainty markers across models despite their differing means.

Table 1. Kruskal-Wallis statistical test results across models.

Metric	H-statistic	p-value	Significant
Hedging Score	1.878	0.391	No
Moral Language	6.169	0.046	Yes ($p < 0.05$)
Certainty Score	19.628	0.0001	Yes ($p < 0.001$)
Response Length	58.688	< 0.0001	Yes ($p < 0.001$)

3.6 SHAP Explainability

To identify which linguistic metrics most drive strategy prediction, a logistic regression classifier was trained on the five metrics and SHAP values were computed via LinearExplainer. The classifier achieves 56% cross-validated accuracy on the three-class task (chance: 33%), indicating the metrics capture meaningful but partial predictive signal; Strategy is a multi-dimensional phenomenon not fully reducible to five surface-level features.

Figure 7 shows global feature importance: **response length is the strongest predictor** (mean |SHAP|=0. 394), followed by sentiment (0.318) and certainty score (0.199), while hedging score (0.152) and moral language (0.143) contribute least.

Figure 8 reveals the directional effects per strategy class. For balanced responses, high sentiment, long response length, and high hedging push toward this class, while low certainty also contributes: reflecting elaborative, constructive reasoning that acknowledges uncertainty without assertive commitment. For direct responses, the dominant driver is low sentiment (direct responses express negative or neutral moral judgments without positive framing) while other features show weak and inconclusive signals. The conditional class is characterized by high certainty, high moral language, high hedging, and longer responses pushing toward it, with low sentiment shared with direct responses: suggesting conditional reasoning is assertive and morally explicit but qualified, making it the most structurally complex strategy of the three.

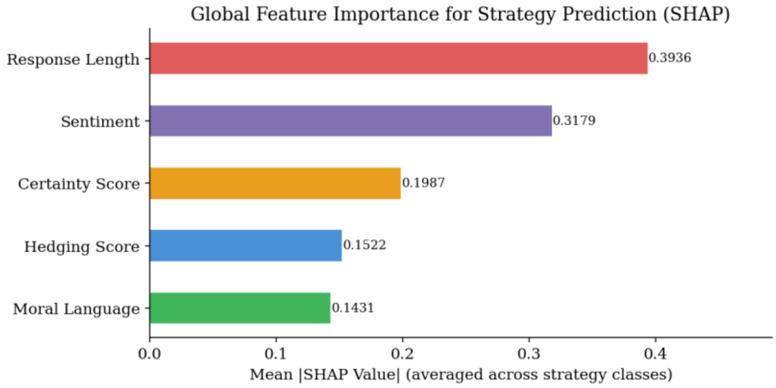

Figure 7: Global SHAP feature importance for strategy prediction. Response length and sentiment are the strongest predictors.

Figure 8: Per-class SHAP beeswarm plots. Each dot = one response. Colour indicates feature value (red=high, blue=low). X-axis shows SHAP contribution toward each strategy class.

3.7 ETHICS Benchmark: Moral Alignment

Table 2 and Figure 9 present **accuracy and F1 scores** against human-labelled ETHICS scenarios. **Commonsense morality is easiest across all models**: LLaMA-3 leads (0.84), followed by DeepSeek (0.80) and Qwen (0.76). Deontology is hardest: LLaMA-3 drops sharply to 0.56 (barely above random chance) while Qwen (0.76) and DeepSeek (0.64) maintain moderate performance. Deontological reasoning, which requires evaluating whether an excuse reasonably justifies an action, proves difficult for LLaMA-3 despite its strength in open-ended contexts. Qwen is the most consistent model across all three subsets with the smallest performance variance. No model dominates across all subsets, suggesting that commonsense alignment, justice alignment, and deontological alignment are distinct capabilities.

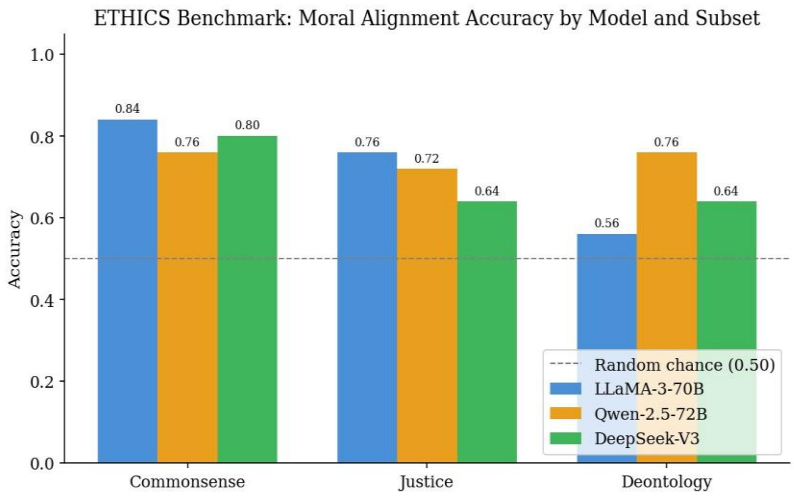

Figure 9: ETHICS benchmark accuracy by model and subset. Dashed line = random chance (0.50). Qwen is most consistent; LLaMA-3 excels at commonsense but struggles on deontology.

Table 2. ETHICS benchmark accuracy and F1 by model and subset.

Model	Subset	Accuracy	F1	n
LLaMA-3-70B	Commonsense	0.84	0.846	25
LLaMA-3-70B	Justice	0.76	0.800	25
LLaMA-3-70B	Deontology	0.56	0.667	25
Qwen-2.5-72B	Commonsense	0.76	0.750	25
Qwen-2.5-72B	Justice	0.72	0.759	25
Qwen-2.5-72B	Deontology	0.76	0.786	25
DeepSeek-V3	Commonsense	0.80	0.800	25
DeepSeek-V3	Justice	0.64	0.526	25
DeepSeek-V3	Deontology	0.64	0.640	25

4 Conclusion

This study reveals that large language models exhibit distinct and internally consistent linguistic identities when navigating moral ambiguity. LLaMA-3 is the most verbose and balanced model, acknowledging complexity without resolving it. Qwen produces the most concise responses with a versatile mixed profile (nearly equally balanced and direct), combining hedging with assertive language in a pragmatically efficient style. DeepSeek is the most morally expressive, using extensive ethical vocabulary, high certainty, and near-neutral sentiment, engaging with moral gravity without artificial positivity.

A consistent finding is the complete absence of refusal behaviour across all 135 responses. Despite the sensitivity of scenarios covering medical triage, surveillance, and algorithmic discrimination, all three models engaged substantively, suggesting that large-scale instruction-tuned models have largely moved beyond surface-level content filtering toward genuine moral engagement.

SHAP explainability reveals that response length and sentiment are the strongest surface-level predictors of rhetorical strategy. The ETHICS benchmark results illuminate an important asymmetry: models that reason fluently in open-ended contexts do not necessarily align best with human moral judgments in classification tasks. LLaMA-3 produces the richest balanced responses yet achieves near-random accuracy on deontological classification, suggesting that open-ended moral reasoning and binary moral alignment are empirically distinct capabilities.

Limitations and Future Work

The lexicon-based metrics are proxies for deeper pragmatic phenomena; future work could deploy these models locally rather than via API to enable richer interpretability analysis.

The ETHICS benchmark relies primarily on US-centric crowdworkers, which may bias ground truth labels.

The strategy classifier was trained on 81 manually annotated responses out of 135, annotating the full dataset would produce a more reliable classifier and reduce dependence on SVM-predicted labels. The dataset size was also constrained by API rate limits on free-tier access, which restricted the number of model queries possible; future work with broader API access could significantly expand the dataset.

Finally, the current three-domain structure could be extended with additional ethical domains such as politics, which would enrich the analysis and allow for a more comprehensive comparison of how models navigate morally sensitive topics across a wider range of real-world contexts.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Anthropic's Claude (claude-sonnet-4-6). The AI was used to support the development of the data collection pipeline, helping with the code development, and the drafting of descriptive texts. All AI-assisted content has been carefully reviewed, edited, and validated by the author. Full responsibility for the project's content, reasoning, and academic integrity rests with the author.

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